

Time: 3 hours

- Note:** 1. Question paper consists of two parts (Part A and Part B)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART - A			10 X 1 = 10 M		
			CO	BTL	Marks
I.	a)	Classify the types of cement	CO1	L2	1 M
	b)	Define True Bearing	CO2	L1	1 M
	c)	What are the specifications of water as per IS: 10500	CO3	L1	1 M
	d)	List the different types of train gauges	CO3	L2	1 M
	e)	What is transpiration	CO3	L1	1 M
	f)	What are the various disciplines of mechanical engineering.	CO1	L1	1 M
	g)	Define Composite material.	CO1	L1	1 M
	h)	Define Casting Process.	CO2	L1	1 M
	i)	What is refrigeration.	CO2	L1	1 M
	j)	List out the various power plants.	CO3	L2	1 M
PART - B			6 X 10 = 60 M		
UNIT - I					
1.	a)	Illustrate the material about steel.	CO1	L2	5 M
	b)	Explain the scope of following disciplines (i) Transportaion Engineering (ii) Water Resource Engineering	CO1	L2	5 M
(OR)					
2.	a)	Write about preparation of bricks.	CO1	L1	5 M
	b)	Explain advantages and disadvantages of prefabricated construction techniques.	CO1	L2	5 M
UNIT - II					
3.	a)	Define surveying . Explain the principles of surveying.	CO2	L1	5 M
	b)	Convert the following into RB. (i) $48^{\circ} 15'$ (ii) $325^{\circ} 30'$ (iii) 175°	CO2	L4	5 M
(OR)					
4.	a)	The following readings are successively taken with a level 0.675 , 1.23 , 0.75 , 2.565 , 2.225 , 1.935 , 1.835 , and 3.22 . The instrument is shifted after the 2 nd and 5 th readings. Prepare a level book and Calculate all reduced levels of different points. The RL of the 1 st point is 100.00m . Use Rise and fall method.	CO2	L4	10 M
UNIT - III					
5.	a)	Explain about Hydrology.	CO3	L2	5 M
	b)	Expalin about types of highway pavements.	CO3	L2	5 M
(OR)					
6.	a)	Explain the components of dam with neat sketch.	CO3	L2	5 M
	b)	List and explain various component parts of Harbour.	CO3	L1	5 M
UNIT - IV					
7.	a)	Explain the role of Mechanical engineering in following sectors (i) Energy (ii) Manufacturing	CO1	L2	5 M
	b)	What are smart materials? list out their properties and applications.	CO1	L2	5 M

(OR)

Explain the contributions of Mechanical Engineering to the welfare of society?

CO1

L2

5 M

What are ceramic materials? list out their properties and applications.

CO1

L2

5 M

UNIT - V

a) Define forming process? Explain about any three forming processes in detail.

CO2

L2

5 M

b) Differentiate between the two stroke and four stroke engines.

CO2

L2

5 M

(OR)

10. a) Explain the working of four stroke SI engine with the help of a neat sketch.

CO2

L2

5 M

b) Explain about the refrigeration cycle with a neat sketch.

CO2

L2

5 M

UNIT - VI

11. a) Explain the steam power plant with a neat sketch.

CO3

L2

5 M

b) What is meant by power transmission drives? Explain any two drives in detail.

CO3

L2

5 M

(OR)

12. a) Explain the diesel power plant with a neat sketch.

CO3

L2

5 M

b) Explain about the applications of robotics in various sectors.

CO3

L2

5 M

***** End of the Question Paper *****

B. Tech II Semester Regular Examinations
BASIC CIVIL & MECHANICAL ENGINEERING
 (CE,EEE,ME,CSE,ECE)
 (Model Paper-II)

Max Marks: 70M

Time: 3 hours

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART - A		10 X 1 = 10 M			
		CO	BTL	Marks	
a)	Classify the types of bricks.	CO1	L2	1 M	
b)	What are the objectives of surveying .	CO2	L1	1 M	
c)	Define Hydrology.	CO3	L1	1 M	
d)	What are the components of harbour.	CO3	L1	1 M	
e)	Clasify the types of cement concrete.	CO1	L2	1 M	
f)	What are the smart materials.	CO1	L1	1 M	
g)	Define joining process.	CO2	L1	1 M	
h)	What are the four strokes of an engine.	CO2	L1	1 M	
i)	Define air conditioning.	CO3	L1	1 M	
j)	What are the applications of gear drives.	CO3	L1	1 M	
PART - B		6 X 10 = 60 M			
UNIT - I					
1.	a)	Write a short notes on scope of civil engineering.	CO1	L2	5 M
	b)	Illustrate the material about aggregate.	CO1	L2	5 M
(OR)					
2.	a)	Describe the role of civil engineers in society.	CO1	L1	5 M
	b)	Explain about manufacturing process of cement.	CO1	L2	5 M
UNIT - II					
3.	a)	What are the levelling instruments used for levelling and explain about any one of the instrument.	CO2	L1	5 M
	b)	Convert the following into whole circle bearing (i) N 39 ⁰ 45' E (ii) S 79 ⁰ 50' W (iii) S 82 ⁰ 30 E	CO2	L4	5 M
(OR)					
4.	a)	The following consecutive readings were observed with a levelling instrument. The instrument was shifted after 5th and 11th readings: 0.585, 1.010, 1.735, 3.295, 3.775, 0.350, 1.300, 1.795, 2.575, 3.375, 3.895, 1.735, 0.635 and 0.605. Determine the R.L of various points, if the R.L of the point on which the first Reading was taken is 146.440 use the height of instrument method.	CO2	L4	10 M
UNIT - III					
5.	a)	Discuss the drinking water quality standards as per IS.	CO3	L2	5 M
	b)	Explain the advantages and disadvantages of flexible pavements.	CO3	L2	5 M
(OR)					
6.	a)	Discuss the concept of rainwater harvesting and its significance.	CO3	L2	5 M
	b)	List and explain various component parts of airport.	CO3	L1	5 M
UNIT - IV					
7.	a)	Explain the role of Mechanical engineering in following sectors (i)Automotive (ii) Marine.	CO1	L2	5 M
	b)	What are composite materials? list out their properties and applications.	CO1	L2	5 M

				(OR)		
		Explain the roles of mechanical engineering in industries and society.	CO1	L2	5 M	
		Explain briefly about the ferrous and non-ferrous materials?	CO1	L2	5 M	
UNIT - V						
a)		Explain about the following forming processes. (a) Rolling (b) forging.	CO2	L2	5 M	
b)		Explain the working of four stroke CI engine with the help of a neat sketch.	CO2	L2	5 M	
(OR)						
a)		What is the joining process in manufacturing? Explain any two joining process in detail.	CO2	L2	5 M	
b)		Explain about the components of electric vehicles.	CO2	L2	5 M	
UNIT - VI						
a)		Explain the hydro electric power plant with a neat sketch.	CO3	L2	5 M	
b)		Explain about the following power transmission drives. (a) Belt drive (b) Chain drive	CO3	L2	5 M	
(OR)						
12. a)		Explain the nuclear power plant with a neat sketch.	CO3	L2	5 M	
b)		What are the different types of configurations in robots and explain any two configurations with examples.	CO3	L2	5 M	

***** End of the Question Paper *****

Date: 14 FEB 2024

Regd. No.

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MIC23

Code: 23AD/AM/IT1T05

B. Tech I Semester Regular Examinations, Jan-2024

BASIC CIVIL & MECHANICAL ENGINEERING

(AI&DS, AI&ML, IT)

Max Marks: 70M

Time: 3 hours

- Note:** 1. Question paper consists of two parts (Part A and Part B)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART - A		10 X 1 = 10 M		
PART-A		Marks	BTL	CO
	What are the various disciplines of civil engineering.	1M	L1	CO1
	Define Contour mapping	1M	L1	CO2
	Classify the types of pavements	1M	L2	CO3
	Classify the types of sources of water.	1M	L2	CO3
	Classify the types of cement	1M	L2	CO1
PART-B				
	What are ferrous materials? list any two properties.	1M	L1	CO4
	Define heat engine. What are the types of heat engines.	1M	L2	CO5
	Define casting in manufacturing process.	1M	L2	CO4
	What is the power plant? List out any two power plants.	1M	L2	CO6
	Define configuration in robots.	1M	L1	CO6
PART - B		6 X 10 = 60 M		
PART-A				
UNIT - I				
a)	Write short notes on scope of Civil Engineering.	5M	L1	CO1
b)	Write short notes on safety of Civil Engineering.	5M	L1	CO1
(OR)				
a)	Explain about Chemical Composition of cement and classify the types of cement.	5M	L2	CO1
b)	Classify the types of aggregates and explain in detail.	5M	L2	CO1
UNIT - II				
a)	Write the objectives and classify and explain the surveying based on instruments.	5M	L1	CO2
b)	Define surveying. Explain the principles of surveying.	5M	L1	CO2
(OR)				
4.	The following readings were observed successfully with a levelling instrument. The instrument was changed after 5th and 11th readings 0.585, 1.010, 1.735, 3.295, 3.775, 0.350, 1.300, 1.795, 2.575, 3.375, 3.895, 1.735, 0.635, 1.605. Draw up a page of level field book and determine the RL of various points if the point on which the first reading was taken is 136.440. Apply the rise and fall method.	10M	L4	CO2
UNIT - III				
a)	Explain the importance of transportation in nation's economic development.	5M	L4	CO3
b)	What are the components of a railway track and explain in detail	5M	L4	CO3
(OR)				
a)	Explain the components of dam with neat sketch.	5M	L2	CO3
b)	Explain the water quality parameters.	5M	L2	CO3

PART-B**UNIT - IV**

Explain the roles of mechanical engineering in industries and society	5M	L2	CO4
What are composite materials? list out their properties and applications	5M	L2	CO4

(OR)

Explain the contributions of Mechanical Engineering to the welfare of society?	10M	L2	CO4
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UNIT - V

Explain about the following forming processes. (a) Rolling (b) forging	5M	L2	CO5
Differentiate between the spark ignition (SI) and compression ignition (CI) engines?	5M	L2	CO5

(OR)

What are the various CNC elements explain in detail with proper sketch.	10M	L2	CO5
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UNIT - VI

a) Explain the steam power plant with neat sketch.	5M	L2	CO6
b) Explain about the following power transmission drives. (a) Belt drive (b) Chain drive (c) Rope drive	5M	L2	CO6
Describe about Robot configurations with neat sketches?	10M	L2	CO6

***** End of the Question Paper *****

QUESTION BANK

UNIT-I

Basics of Civil Engineering

Define Civil Engineering and explain roles and responsibility of civil engineering.

Write various disciplines in Civil Engineering?

Explain briefly about scope of various engineering disciplines

Structural engineering

i) Geo-technical engineering

ii) Transportation engineering

iv) Hydraulic and water resources engineering

(v) Environmental engineering

4. Define cement and explain chemical composition of cement?

5. Explain field and laboratory tests can be done on cement?

6. Write about different types of cements?

7. Explain about manufacturing process of cement by dry & wet process.

8. Define brick and briefly explain its composition.

9. Write the characteristics of a good brick and explain about manufacturing process of cement.

10. Differentiate Clamp burning & Klin burning?

11. Explain various bonds in brick work with the help of neat sketch.

12. Explain steps during construction

13. Define concrete and its importance and write any five advantages and disadvantages of concrete?

14. Define aggregate and briefly explain classification of aggregate?

15. Write the properties and various tests to be conducted on aggregate?

16. Explain manufacturing process of Concrete?

17. Write the elements of a building?

18. Define steel and explain various types of steel?

19. Define prefabrication and explain about advantages and disadvantages of prefabrication?

20. Explain briefly about materials used in prefabrication.

21. Explain briefly about types of prefabrication.

UNIT-II

Surveying

- Define Surveying and its importance
- What are the main objectives and principle of surveying?
- Differentiate plane and geodetic surveying?
- Explain about classification of surveying
- Classification Based Upon the Nature of the Field
- i) Classification Based Upon the Objective of Survey
- ii) Classification Based Upon Methods Employed
- iv) Classification Based Upon the Instruments
5. Explain about horizontal measurement and their methods?
6. Explain about Angular measurement and their methods?
7. Define compass and types of compass
8. Briefly explain various instruments used in surveying>
9. Differentiate prismatic compass and surveyor's Compass
10. Define
- (i) True bearing
- (ii) Magnetic Bearing
- (iii) Whole circle bearing
- (iv) Reduced ,fore and back bearing
11. Define levelling and its classification.
12. Comparison of Height of Collimation Method with the Rise and Fall Method.
13. Define contouring and its charcterstics
14. What are the uses of contour map
15. Convert the following WCB to RB
1. $20^{\circ} 30'$ 2. $176^{\circ} 10'$ 3. $220^{\circ} 30'$ 4. $295^{\circ} 40'$
16. The WCB of a line is (i) $N 25^{\circ} 30' E$ (ii) $S 65^{\circ} 10' E$ (iii) $S 32^{\circ} 30' W$ (iv) $N 40^{\circ} 20' W$. What will be the RB in each case?
17. The following are the observed fore bearings of the traverse sides: AB, $80^{\circ} 30'$ BC, $150^{\circ} 15'$; CD, $270^{\circ} 20'$ and DE, $325^{\circ} 30'$. Find their back bearings.
18. The FB of the lines are as follows:
AB, $N 40^{\circ} 45' E$; BC, $S 80^{\circ} 20' E$; CD, $S 50^{\circ} 30' W$, DE, $N 70^{\circ} 15' W$. Find their back bearings.
19. Convert the following WCBs to RBs
WCB of AB = $45^{\circ} 30'$
WCB of BC = $125^{\circ} 30'$
WCB of CD = $222^{\circ} 45'$
WCB of DE = $325^{\circ} 15'$
20. Convert the following RBs to WCBs
RB of AB = $S 35^{\circ} 30' W$
RB of BC = $N 25^{\circ} 30' E$
RB of CD = $N 62^{\circ} 45' W$
RB of DE = $S 25^{\circ} 15' E$

UNIT-III

Transportation Engineering & Water Resources and Environmental Engineering

- What is a pavement and types?
- What is rigid pavement?
- List any two differences between flexible and rigid pavements.
- Define harbor?
- What are the requirements of a harbor?
- What is a tunnel and its uses?
- What is flexible pavement?
- What is the permissible limit of chlorine?
- List out any two examples for surface and sub-surface sources of water?
- What is environmental engineering?
- Define hydrological cycle.
- Define rainwater harvesting.
- What is a dam and its types?
- What is a reservoir and its types?
- Explain the importance of transportation in nation's economic development?
- Distinguish between flexible and rigid pavements?
- What are the components and classification of harbors?
- Explain about the tunnel and its components
- Explain about the airport and its components
- What is a permanent way and what are its components?
- What are the sources of water?
- Explain about the hydrology
- Write any 10 permissible limits of water as per IS 10500:2012
- Explain about the rainwater harvesting.
- Write a short note on water storage and conveyance structures?